Synopsis

Loan Default Prediction Using Machine Learning Techniques

Problem Id: 2

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Loan Default Prediction using Machine Learning

1. Problem Statement

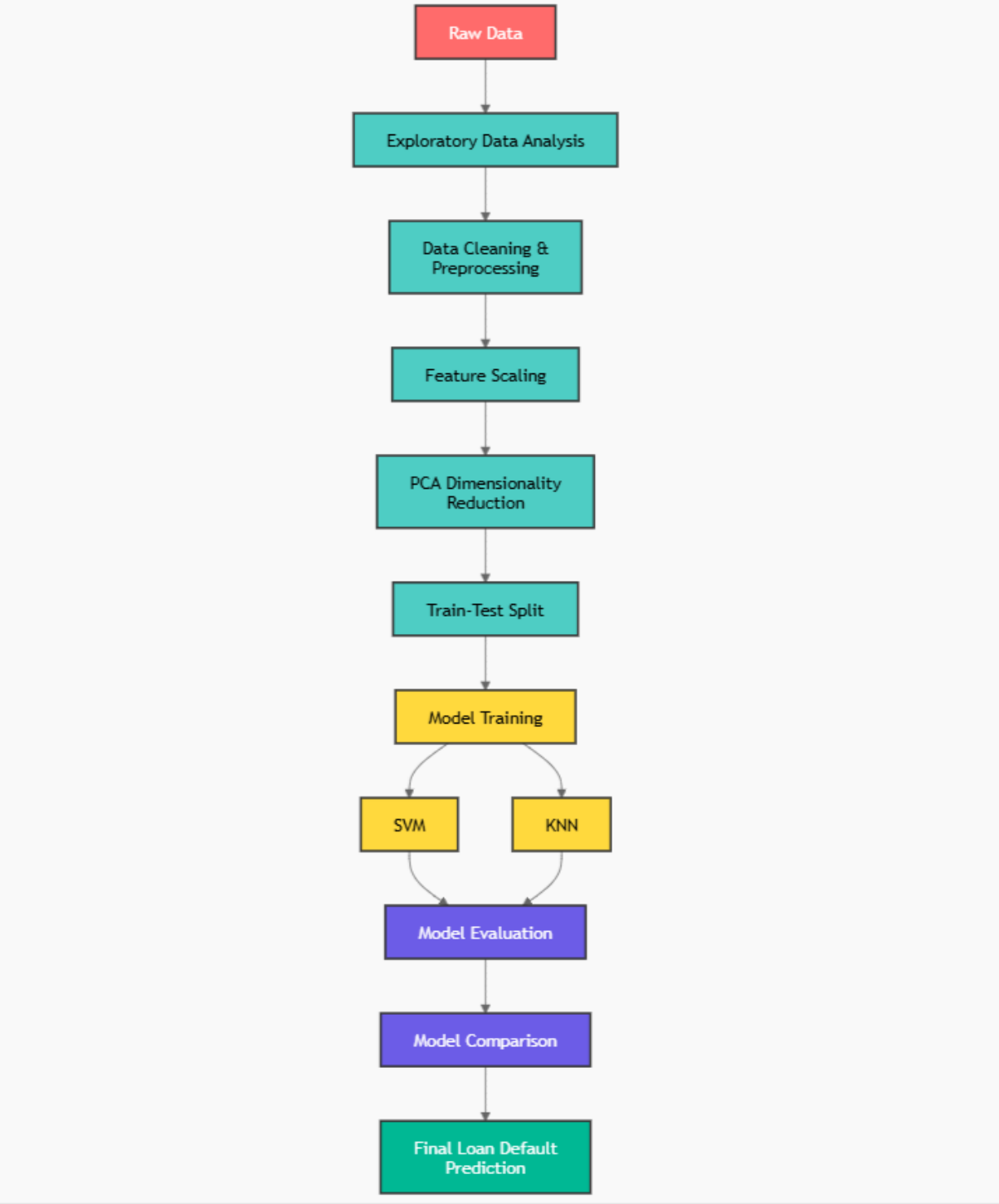
The problem focuses on building a machine learning classification system that predicts whether a customer is likely to default on a loan using banking data. The process begins with Exploratory Data Analysis (EDA), where the dataset is examined to understand patterns, relationships between variables, and data distribution. After that, data preprocessing is performed to clean missing or inconsistent values, normalize numerical features, and transform the data for better model performance.

Since banking datasets often contain many variables, Principal Component Analysis (PCA) is applied to reduce dimensionality while preserving important information. The system then trains classification models such as Support Vector Machine (SVM) with different kernel functions (linear, polynomial, and RBF) and K-Nearest Neighbors (K-NN) to identify patterns that indicate loan default risk.

The trained models are evaluated using performance metrics like precision, recall, F1-score, and ROC–AUC, which measure how accurately the system predicts defaults.

In real life, such systems help banks and financial institutions assess credit risk, approve loans more responsibly, reduce financial losses, and make data-driven lending decisions. This improves financial stability and ensures that loans are provided to customers who are more likely to repay them

2.System Workflow Diagram (Fig. 1)



3. Explanation of Each Step

3.1 Dataset Collection

Banking data is collected, which may include:

- Customer income
- Loan amount
- Credit history
- Employment status

3.2 Exploratory Data Analysis (EDA)

EDA is used to understand the dataset by:

- Visualizing distributions (histograms)
- Finding relationships (correlation heatmaps)
- Detecting outliers (box plots)

Purpose:

- Identify patterns
- Detect anomalies
- Understand feature importance

3.3 Data Preprocessing

This step prepares data for modeling:

- Handling missing values
- Encoding categorical variables
- Feature scaling (normalization/standardization)

Why important?

Machine learning models perform better on clean and properly scaled data.

3.4 Dimensionality Reduction (PCA)

Principal Component Analysis (PCA) reduces the number of features while preserving important information.

Advantages:

- Reduces computation time
- Avoids overfitting
- Improves model performance

3.5 Model Training

(A) Support Vector Machine (SVM)

SVM finds the optimal boundary (hyperplane) to separate classes.

Kernels used:

- Linear Kernel → for linearly separable data
- Polynomial Kernel → for curved boundaries
- RBF Kernel → for complex non-linear patterns

(B) K-Nearest Neighbors (K-NN)

K-NN classifies a data point based on the majority class of its nearest neighbors.

4. Why These Models Are Used

Why SVM?

- Works well in high-dimensional data
- Effective for complex classification problems
- Kernel trick allows handling non-linear data

Why K-NN?

- Simple and easy to understand
- No training phase (lazy learner)
- Works well for smaller datasets

5. Why Other Models Are Not Preferred

Decision Trees

- Can overfit easily on complex datasets

Random Forest

- More computationally expensive
- Less interpretable compared to SVM

Logistic Regression

- Works only well for linear relationships
- Not suitable for complex patterns in loan data

Neural Networks

- Require large datasets
- High computational cost
- More complex to implement

6. Model Evaluation Metrics

Accuracy

Measures overall correctness of predictions.

$$\boxed{\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}}$$

Precision

Measures how many predicted defaulters are actually defaulters.

$$\boxed{\text{Precision} = \frac{TP}{TP + FP}}$$

Recall

Measures how many actual defaulters are correctly identified.

$$\boxed{\text{Recall} = \frac{TP}{TP + FN}}$$

F1-Score

Harmonic mean of precision and recall.

$$\text{F1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

ROC-AUC

Measures model's ability to distinguish between classes.

True Positive Rate (TPR) vs False Positive Rate (FPR)

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

7. Visualization Techniques

- **Histogram** → Distribution of numerical data
- **Box Plot** → Outlier detection
- **Heatmap** → Feature correlation

8. Libraries Used

- NumPy → Numerical operations
- Pandas → Data handling
- Matplotlib → Basic plots
- Seaborn → Advanced visualization
- Scikit-learn → Machine learning models

9. Required Topics

1. Exploratory Data Analysis (EDA)

EDA is the process of analyzing and understanding the dataset before applying machine learning.

2.Data Preprocessing

Data preprocessing prepares raw data into a clean and usable format for machine learning models.

3. Dimensionality Reduction (PCA)

Principal Component Analysis (PCA) is used to reduce the number of features while retaining important information.

4. Support Vector Machine (SVM)

SVM is a supervised learning algorithm used for classification.

5. K-Nearest Neighbors (KNN)

KNN is a simple, instance-based learning algorithm.

10. Platform Used

- Jupyter Notebook /
  Visual Studio Code /Python IDLE

11. Conclusion

This system helps financial institutions:

- Predict loan defaults accurately
- Reduce financial risk
- Make data-driven decisions

By combining preprocessing, PCA, and powerful models like SVM and K-NN, the system achieves efficient and reliable predictions.

12. References

- Scikit-learn Documentation: https://scikit-learn.org
- https://scikit-learn.org/stable/modules/preprocessing.html
- https://arxiv.org/abs/1907.01800